



Portfolio Summary

Katherine Stanton



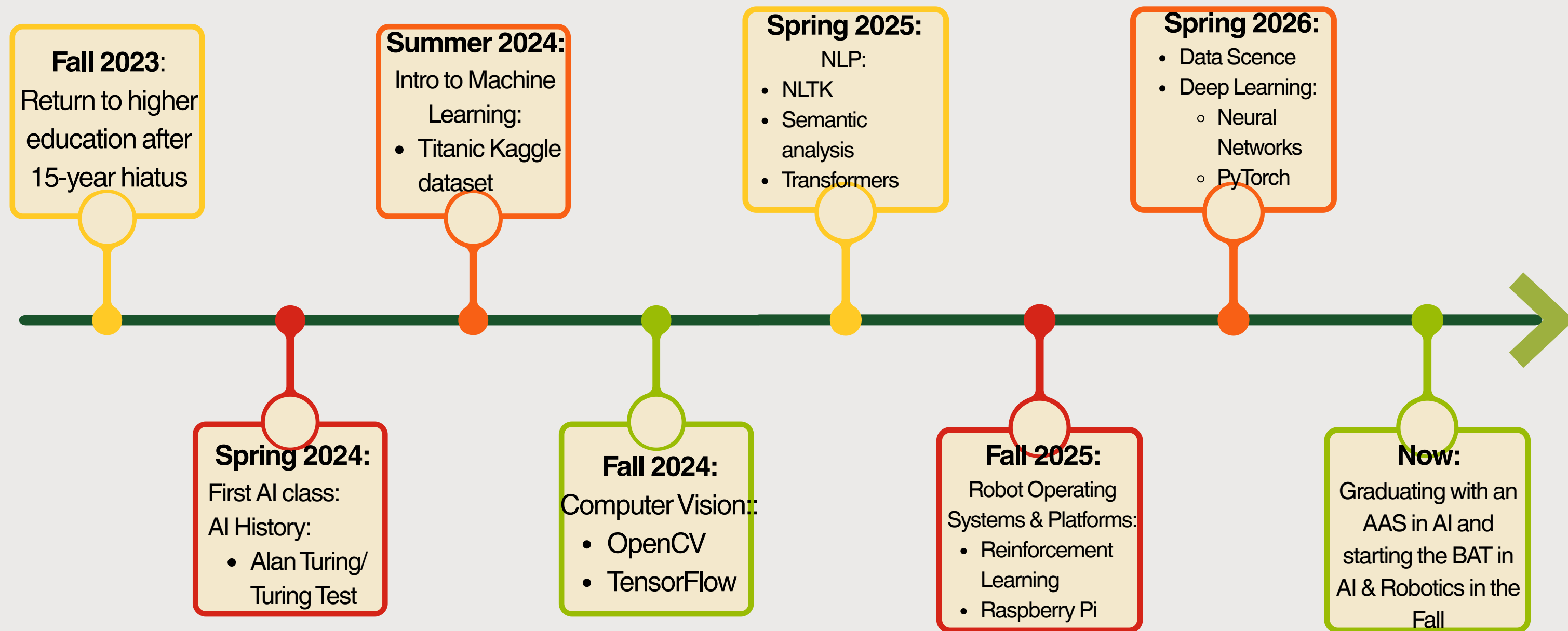
AAS Artificial Intelligence Program at Houston City College

Student entering her Junior year with an interest in Computer Vision, NLP, Automation and ROS

GitHub repository: <https://github.com/ktgraze/Katherine-Stanton-AI-Portfolio.git>

LEARNING JOURNEY TIMELINE

KATHERINE STANTON
AAS IN AI PORTFOLIO



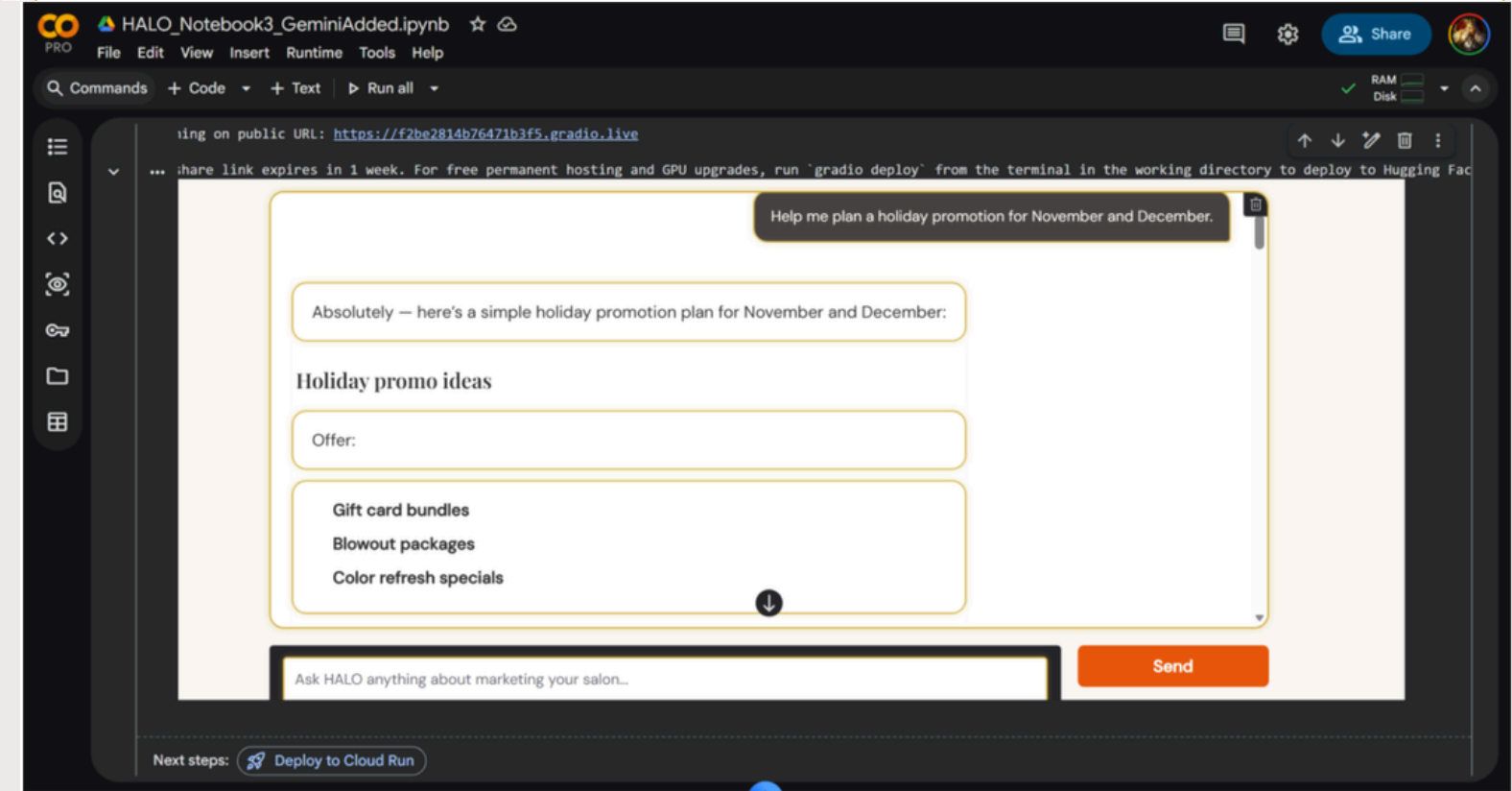
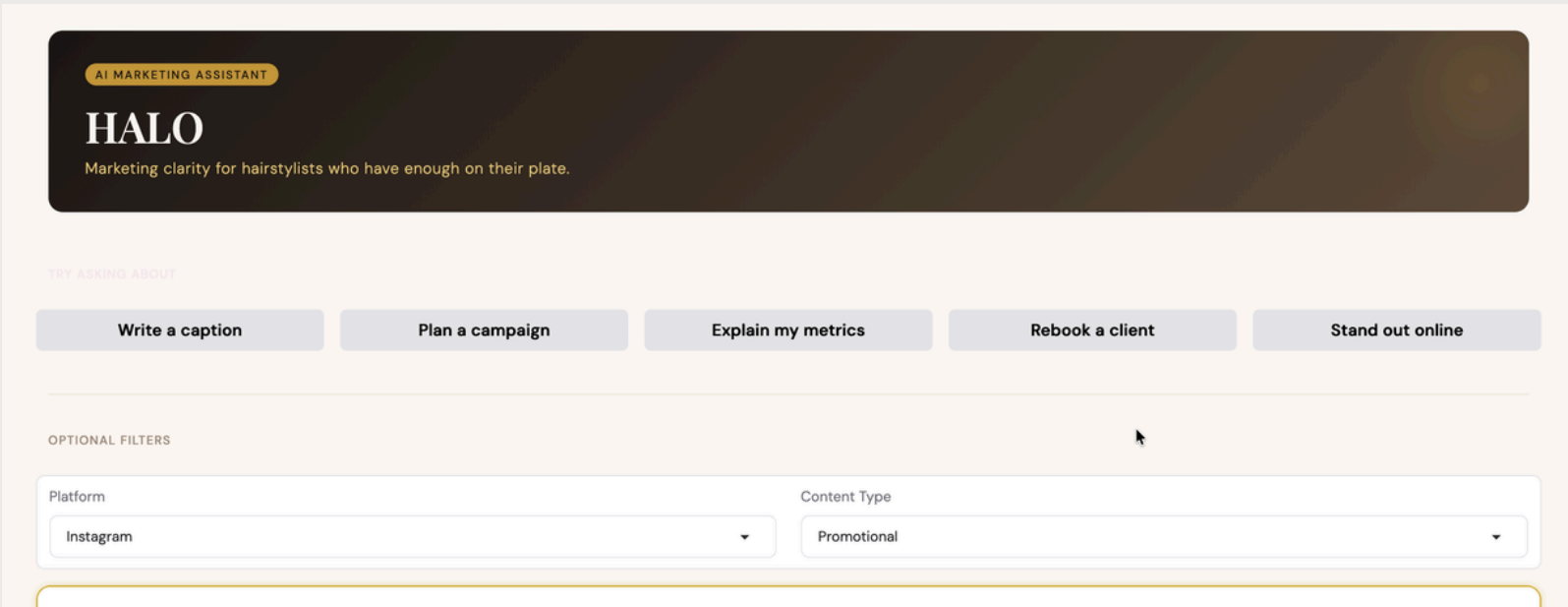
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Featured Projects

HALO: Marketing Assistant for Hairstylists

HALO is a domain-specific Retrieval-Augmented Generation system designed for the beauty and hair industry

GitHub Repository [Link](#)

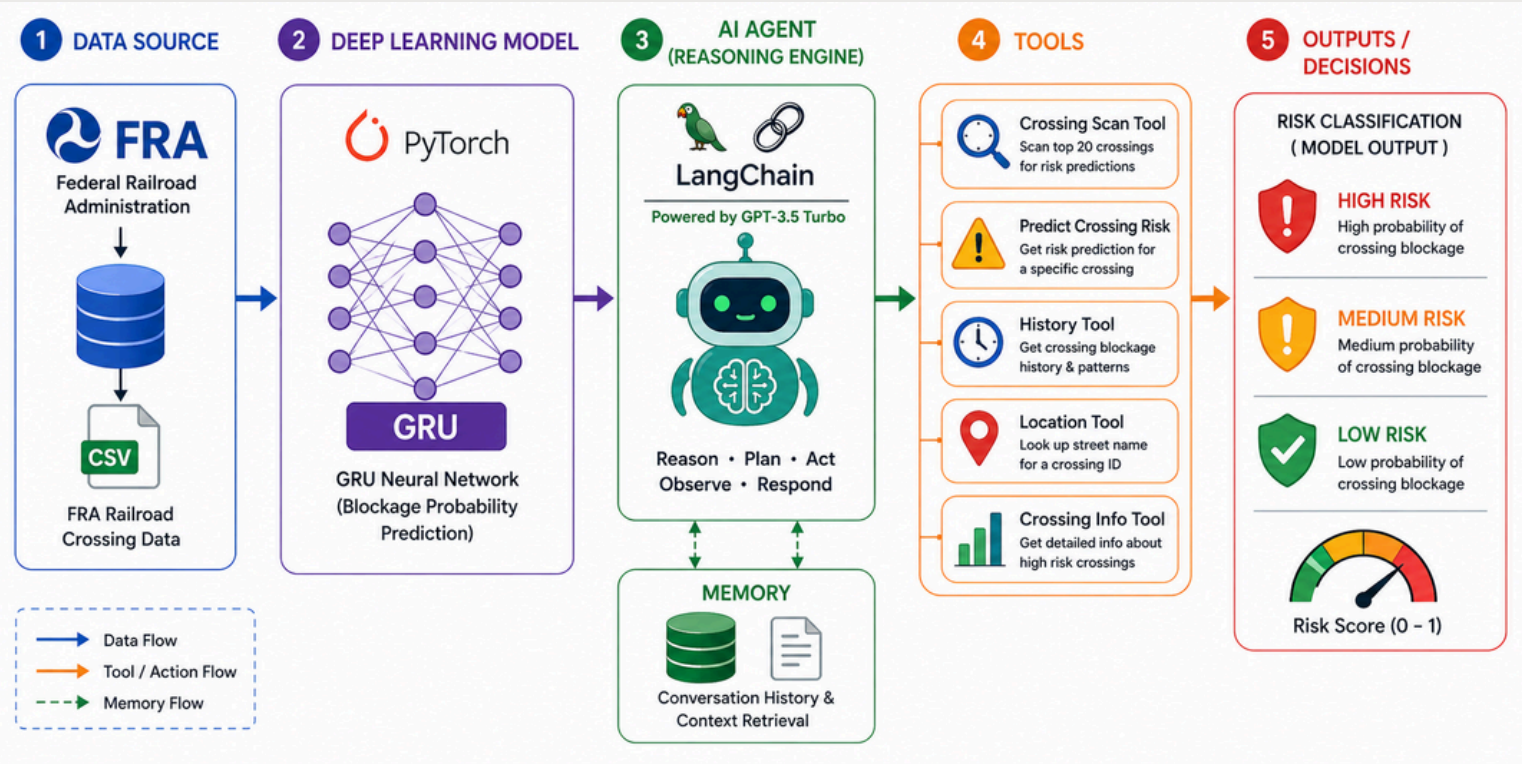


Houston Railroad Crossing Blockage Prediction

Agent

LangChain agent that uses historical FRA data to predict the probability that a railroad cross in Houston, Texas will be blocked.

GitHub Repository [Link](#)



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SCENARIO 2 – Multi-Crossing Risk Scan
=====
A fire station commander wants a risk overview
of all major crossings at 8am on a Monday in April.

> Entering new AgentExecutor chain...
I should use the ScanHighRiskCrossings tool to identify the high-risk crossings at this specific time.
Action: ScanHighRiskCrossings
Action Input: 8, 0, 4CROSSING RISK SCAN – 8:00 on Monday
Scanning top 20 most historically blocked crossings

HIGH RISK (1 crossings):
EASTWOOD STREET (859522Y) – 100.0%

MEDIUM RISK (12 crossings):
Leeland Street (288224V) – 42.6%
Cullen Boulevard (288221A) – 46.1%
Milby Street (288226J) – 55.9%
LOCKWOOD STREET (859523F) – 73.3%
DUMBLE STREET (859524H) – 78.8%
McKinney Street (288227R) – 73.9%
OAKHURST STREET (859521S) – 66.4%
MILBY STREET (859518J) – 65.5%
LAMDALE STREET (288050B) – 65.4%
Hirsch Road (755640L) – 62.5%
York Street (288228X) – 68.0%
Sampson Street (288229E) – 56.3%
The Eastwood Street crossing (ID: 859522Y) is currently at high risk of blockage at 8am on Monday. It is crucial to be aware of this when plann.
Final Answer: The Eastwood Street railroad crossing is currently at high risk of blockage at 8am on Monday in April.

> Finished chain.

=====
FINAL ANSWER:
=====
The Eastwood Street railroad crossing is currently at high
risk of blockage at 8am on Monday in April.
=====
```

Skills & Competencies

Frameworks

Scikit-learn

TensorFlow Keras

PyTorch 2.3.0

LangChain 0.2.16

Core Tools

Google Colab GPU

Python 3.12

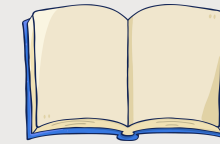
Pandas | NumPy data

Domains

Computer Vision OpenCV

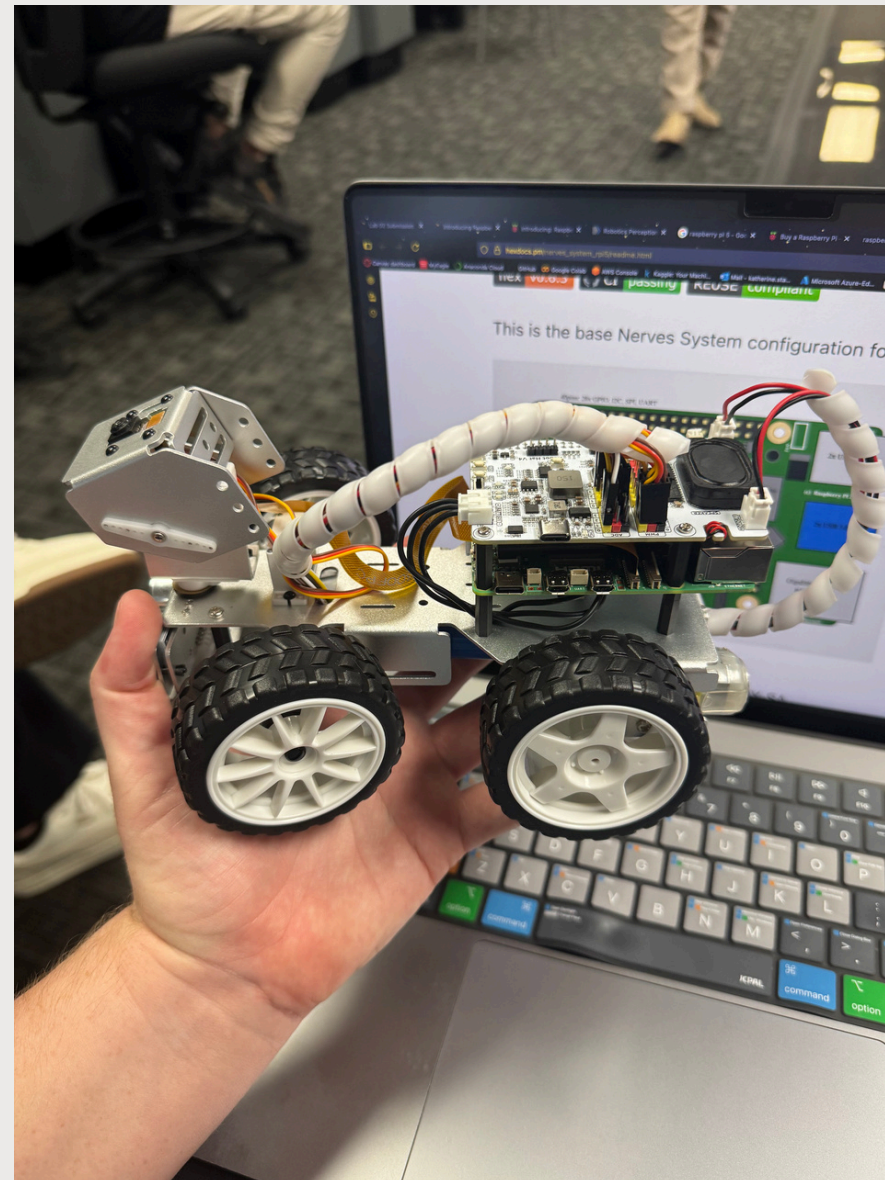
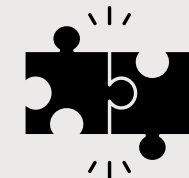
Robotics ROS

NLP HuggingFace



Soft Skills

- Independent Research
- Adaptability
- Self-Directed Learning
- Research Mindset



Reflection & Next Steps



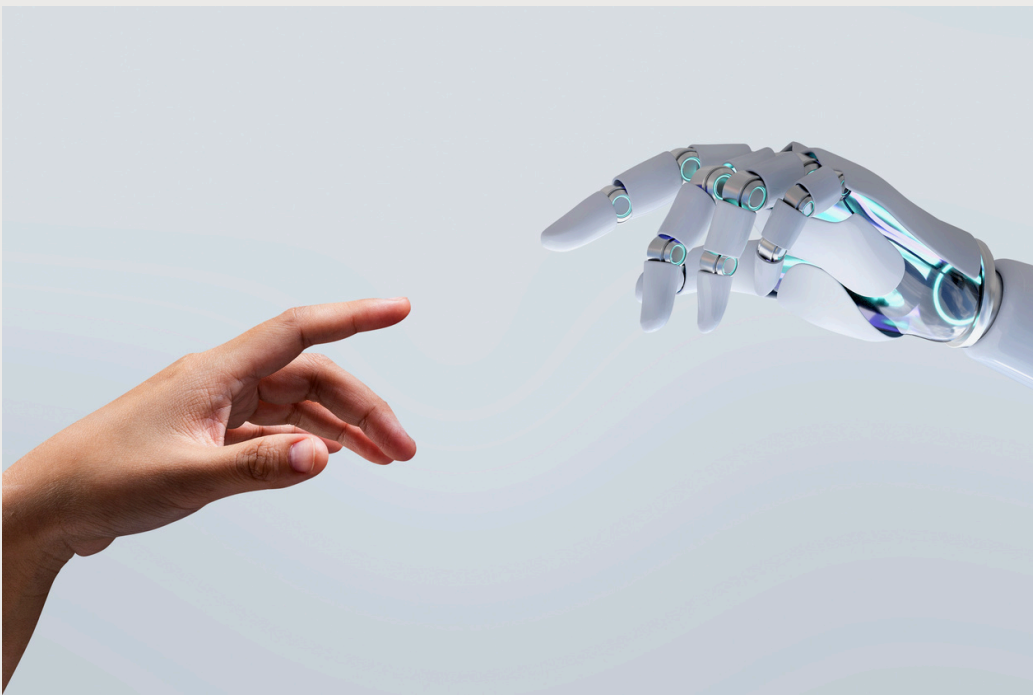
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[LinkedIn](#)

Reflection

Starting from a non-tech background, not knowing any programming languages or how to set up a coding environment, I have come a far way. The courses in the Applied AI program have built confidence in me and piqued my curiosity. From Machine Learning to Computer Vision to Natural Language Processing to Deep Learning. I am looking forward to starting the BAT program!



Next Steps

- AAS Graduation May 2026
- Transition to a job in the tech industry.
- Begin BAT program August 2026
- BAT Graduation May 2028
- Summer projects: Codecademy courses:
 - Statistics for Machine Learning
 - Intro to C++
- DIY ROS2 project
- Building on my Deep Learning Final Project: incorporating more datasets, APIs, and adding a RL component.

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